CHVI for Uganda: Object-Oriented C++ System

Programming Project Write-up

Kuli Faith Rombe (Reg. No. 2025/HD06/22009U)
School of Statistics and Planning, Makerere University

November 2026

1 Project definition

Topic. A Climate–Health Vulnerability Index (CHVI) for Uganda: Construction and Application using Factor Analysis of Mixed Data (FAMD), implemented as an object-oriented, menu-driven C++ system.

1.1 Objectives

The general objective is to construct and apply a household-level CHVI and to examine its association with fever in children under five and with residential disparities. The specific objectives, each with its null hypothesis, are:

1. To construct a household-level CHVI and evaluate its robustness. H01: the CHVI does not exhibit acceptable robustness.
2. To ascertain the association between the CHVI and fever in children under five. H02: the CHVI has no statistically significant association with fever.
3. To ascertain rural–urban disparities in the CHVI. H03: there are no statistically significant rural–urban disparities in the CHVI.

1.2 Background

Climate change is a growing threat to health in low-income countries such as Uganda. Population growth of 3.0% a year, high poverty and reliance on rain-fed smallholder agriculture make household income and well-being sensitive to rainfall and temperature (Isunju et al., 2024; van de Ven et al., 2019). Malaria remains the leading cause of outpatient attendance, at 32.1% of all diagnoses (Isunju et al., 2024).

Vulnerability to climate hazards is conventionally defined by the IPCC framework as a function of exposure (E), sensitivity (S) and adaptive capacity (A), V = f(E, S, A). Children, older people and low-income households carry most of the risk, because they have more existing health problems and fewer resources to cope (Isunju et al., 2024). Uganda enacted a National Climate Change Act in 2021, yet the health sector still lacks a standardized, national measure of climate–health vulnerability (Isunju et al., 2024; Erima et al., 2025).

1.3 Problem statement

Existing Ugandan studies address single sectors or single hazards, such as floods in the Manafwa catchment and no household-level index combines exposure, sensitivity and adaptive capacity for

the whole country (Erima et al., 2025). Building such an index is a statistical problem in its own right, because household surveys mix continuous, ordinal and nominal variables. Classical principal component analysis assumes continuous, Gaussian data and is unsuitable for them (Pagès, 2004). A second problem is validation: composite indices are rarely tested against clinical outcomes (Conlon et al., 2020).

The statistical problems the system must therefore handle are: (i) combining mixed-type variables into one index with objective weights; (ii) showing that the index is stable under reasonable analytic choices; (iii) testing its association with a binary health outcome measured on children nested in survey clusters; and (iv) testing a categorical disparity between rural and urban households.

1.4 Brief literature survey

The IPCC exposure–sensitivity–adaptive-capacity framework underlies most vulnerability indices (Cutter et al., 2003; Vincent, 2004; Zhang et al., 2025). variable-based indices simplify complex socio-environmental conditions and help allocate scarce resources (Abdrabo et al., 2023). The literature clusters around three points.

- **Methods for mixed data.** FAMD generalises PCA and multiple correspondence analysis. It standardises continuous variables and scales the variable columns of categorical variables by category frequency, so both types contribute in a balanced way (Pagès, 2004; Husson et al., 2010).
- **Climate, vulnerability and child health.** In northern Benin, PCA-based malaria vulnerability scores rose with the number of children under five and fell with bed-net use and phone ownership (Gbaguidi et al., 2025). In the Lake Victoria Basin, vegetation loss and rainfall deficits predicted child stunting (López-Carr et al., 2016). Wealth, maternal education and floor material are structural determinants of child disease burden (Obasohan et al., 2023).
- **The gap.** Published vulnerability indices are mostly descriptive maps with little validation against household-level health outcomes (Conlon et al., 2020; Wang et al., 2026). Rural and urban settings also differ structurally. Rural households depend on rain-fed farming while urban informal settlements face flooding and failed sanitation (Erima et al., 2025; Isunju et al., 2026).

This project fills the gap with a validated, household-level CHVI whose association with fever and rural–urban divide are tested within one reproducible program.

1.5 How the project is tackled

The work is delivered as one C++ program of about 3,500 lines with no external libraries. It reads the UDHS 2022 Kids Recode (Stata .dta file) and 32 years of daily weather data, recodes the raw variables, builds the CHVI with FAMD, runs three robustness analyses, fits multilevel logistic models and tests residence disparities. Object-oriented design (classes, inheritance, polymorphism, encapsulation) organises the statistics and data structures (hash maps, heaps, sorted vectors with binary search, queues) carry the data.

2 Statistical theory and planned approach

The program turns 16 household variables and daily weather records into one standardized index, then tests it. This section gives the theory behind each step, the equation numbers are the ones used in the code comments.

2.1 Data

The survey data are a 10% extract of the UDHS 2022 Kids Recode (Uganda Bureau of Statistics and ICF, 2023) 1,436 child records from 995 households in 586 enumeration areas (clusters). Climate data are 168,029 daily records for 14 regions, 1991–2022, from Open-Meteo (Zippenfenig, 2024) temperature at 2 m, relative humidity, precipitation and wind speed at 10 m. Exact duplicate rows (4,380) are dropped first.

2.2 Climate exposure

Daily values are reduced to one value per region and day, then to annual values mean temperature, humidity and wind and total precipitation. Exposure is the standardized anomaly of 2021–2022 against the 1991–2020 baseline, for region r and climate variable v:

$$z_{r,v} = \frac{\bar{X}_{r,v}^{(2021\text{–}2022)} - \bar{X}_{r,v}^{(1991\text{–}2020)}}{\sigma_{r,v}^{(1991\text{–}2020)}} \tag{2}$$

Every household inherits the four z-scores of its region.

2.3 variables and scoring

Sixteen variables follow the IPCC framework. Each has a normalized score in [0, 1] oriented in the direction of its domain: higher exposure or sensitivity means more vulnerable and higher adaptive score means more capacity.

Domain	**variables (type)**	**Score**
Exposure (4)	temperature, precipitation, humidity, wind anomalies (continuous)	min–max of the z-score, Eq. (3)
Sensitivity (4)	cooking fuel, roof, floor, wall (nominal)	expert category scores, e.g. roof: natural 1.0, iron sheets 0.5, modern 0
Adaptive capacity (8)	wealth quintile, education, water source, time to water, toilet, phone, bank account, land (ordinal / nominal)	category scores; ordinal codes scaled to 0–1

For a continuous variable the min–max normalization is Eq. (3) and its reversed form Eq. (4) applies where a larger raw value means less of the domain quantity:

$$Y_{ij} = \frac{X_{ij} - \min(X_{\cdot j})}{\max(X_{\cdot j}) - \min(X_{\cdot j})} \tag{3}$$

$$Y_{ij} = \frac{\max(X_{\cdot j}) - X_{ij}}{\max(X_{\cdot j}) - \min(X_{\cdot j})} \tag{4}$$

Categories with no score (one household with an unknown toilet type) take the column median.

2.4 Missing data

The two variables with blanks are water source (2 households) and time to water (211 households). Missing categories are imputed by iterative regularized FAMD with two axes. Starting from the modal category, FAMD is fitted, the variable matrix is reconstructed from the first two axes with shrinkage, each missing cell takes the category with the largest reconstructed value and the cycle repeats until nothing changes.

2.5 Factor Analysis of Mixed Data

FAMD (Pagès, 2004) is a principal component analysis of a combined design matrix Z with n households and J columns. Continuous variables enter standardized each category c of a nominal variable enters as a variable column scaled by its proportion p_c:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j} \text{ (continuous)}$$

$$z_{ic} = \frac{\mathbf{1}[x_i = c] - p_c}{\sqrt{p_c}} \text{ (category } c\text{)}$$

The principal axes are the eigenvectors of $C = Z'Z/n$, found with the Jacobi method. Total inertia equals $N_1 + \sum(q-1)$, where N_1 is the number of continuous variables and q the number of categories of each categorical variable. Here it is 4 + 30 = 34 over J = 46 columns. Factor scores are $F = ZV$.

2.6 Weights, dimension indices and the CHVI

The coordinate of a variable on axis k is its squared correlation with the axis (continuous) or its correlation ratio η^2 (categorical). The weight of variable i uses the first K = 4 axes and their eigenvalues λ:

$$W_i = \frac{\sum_{k=1}^{K} L_{ik}\,\lambda_k}{\sum_{k=1}^{K} \lambda_k} \tag{5}$$

Each dimension index is the weighted mean of its normalized variables (Eq. 6). The composite follows the IPCC additive form with equal dimension weights and is standardized to a z-score:

$$I_{d,j} = \frac{\sum_{i \in d} W_i Y_{ij}}{\sum_{i \in d} W_i} \tag{6}$$

$$\text{CHVI}_j = \tfrac{1}{3}E_j + \tfrac{1}{3}S_j - \tfrac{1}{3}A_j \tag{8}$$

$$\text{CHVI}_j^{*} = \frac{\text{CHVI}_j - \mu}{\sigma} \tag{9}$$

Households are classified into five levels with cut-points at -1.5, -0.5, 0.5 and 1.5 on CHVI*: Very Low, Low, Moderate, High and Very High.

2.7 Validity and robustness

Construct validity is the sign and size of the correlation of CHVI* with each dimension positive for E and S, negative for A. Robustness is judged with Spearman rank correlation (ρ) in three analyses and the index is called acceptably robust when every ρ is at least 0.80.

1. **Component retention.** CHVI with K = 4 axes against K = 12.
2. **Cluster bootstrap.** 200 resamples of enumeration areas with replacement; FAMD is refitted on each resample and the new weights score all households.
3. **Dimension weighting.** Clusters are split 50/50. Fever is regressed on E, S and A in the training half, absolute coefficients, normalized to sum to 1, become empirical weights applied to the test half. The split is repeated 100 times.

2.8 Multilevel logistic regression

Fever in the last two weeks for child i in cluster j follows a random-intercept logistic model:

$$\operatorname{logit} P(y_{ij} = 1 \mid x_{ij}, u_j) = x'_{ij}\beta + u_j, \qquad u_j \sim N(0, \sigma^2) \tag{10}$$

The cluster likelihood integral is evaluated by adaptive Gauss–Hermite quadrature with nodes from the Golub–Welsch algorithm and a Newton search for each cluster's mode. One node gives the Laplace approximation which is the default of R's glmer and is the program default. Parameters are found by BFGS and standard errors from the numerical Hessian. Four models are compared by AIC and the intraclass correlation is $\sigma^2/(\sigma^2 + \pi^2/3)$:

Model	Covariates
M0 Null	random intercept only
M1 Contextual	CHVI*, residence, radio
M2 Individual	age, sex, birth order, birth weight category
M3 Full	M1 plus television, plus M2

2.9 Residence disparity

A Pearson chi-square test of independence is applied to the 5×2 table of vulnerability level by residence. The statistic is the sum of $(O - E)^2/E$ over cells with 4 degrees of freedom, a p-value from the regularized incomplete gamma function and Cramér's V as the effect size.

3 System design: OOP and data structures

The program has 18 source files and about 3,500 lines, built from small classes that each own one statistical task. The classes are grouped in four layers that is input, modelling, analysis and interface.

Each statistical problem of Section 1.3 has its own class, so the design follows the research questions:

Statistical problem (method)	Obj.	Classes	OOP and C++ concepts
Mixed continuous, ordinal and nominal variables combined into one weighted index (FAMD, Eqs. 3–9)	1	`Variable` and its subclasses, `VariableSchema`, `FAMD`, `FAMDResult`, `IndexBuilder`	abstraction, inheritance, polymorphism, composition
Missing categories (iterative regularized FAMD)	1	`FAMDImputer`	single responsibility
Stability of the index (Spearman ρ, cluster bootstrap, split weights)	1	`RobustnessAnalyzer`	composition, const references
Binary fever outcome with children nested in clusters (random-intercept logit, Gauss–Hermite quadrature, BFGS)	2	`MixedLogit`	encapsulation, private optimiser
Rural–urban difference in the CHVI classes (chi-square, Cramér's V)	3	`ChiSquareTest`, `VulnerabilityClassifier`	static utility, binary search
Getting from raw files to analysis data (recoding, Eq. 2)	all	`StataReader`, `WeatherProcessor`, `DHSLoader`, `Dataset`	encapsulation, static factory, hash map

3.1 Object-oriented design

Class	Role	OOP concept
`Variable` (abstract)	common interface of a variable: name, domain, `normalise()`	abstraction
`ContinuousVariable`, `CategoricalVariable`	min–max scoring (Eq. 3/4) or category score table	inheritance
`OrdinalVariable`, `NominalVariable`	ordered / unordered categories	inheritance, polymorphism
`VariableSchema`	owns the 16 variables through `unique_ptr`; adding one is a single line	composition, RAII
`Household`, `Child`, `Dataset`	records and the cluster index	encapsulation
`StataReader`	reads the binary .dta file (format 118)	encapsulation
`WeatherProcessor`, `ClimateIndicators`	daily weather to regional z-scores (Eq. 2)	single responsibility
`DHSLoader`	recoding rules of the survey variables	static factory
`FAMDImputer`	iterative FAMD imputation of missing categories	separation of concerns
`FAMD`, `FAMDResult`	design matrix, eigen-decomposition, loadings, weights	encapsulation
`IndexBuilder`, `VulnerabilityClassifier`	Eqs. 6–9 and the five levels	separation of concerns
`RobustnessAnalyzer`	the three sensitivity analyses	composition
`MixedLogit`	random-intercept logistic regression	encapsulation
`ChiSquareTest`	independence test and Cramér's V	static utility
`TopK<T>`, `AuditLog`	min-heap and bounded queue	templates
`ResearchSystem`	owns all data and results; one method per menu option	façade
`SelfTest`	75 verification checks	testability

Polymorphism does the main work in the indicator layer. `IndexBuilder` calls `normalise()` through a `Variable&` and never asks whether a variable is continuous or categorical, so a new variable type needs only a new subclass. Constructors use initialiser lists, accessors are `const` and errors are reported with exceptions that the menu catches and prints.

3.2 Data structures

Structure	Used for	Why
Hash map (`unordered_map`)	cluster id to household positions; region aggregation; (region, day) accumulators in the weather step	O(1) lookup while resampling clusters and joining children to households
Min-heap (`TopK<T>`)	the k most vulnerable households	O(n log k) instead of a full sort
Sorted vector + binary search	the four classification cut-points	O(log k) per household with `upper_bound`
Dense matrix (`Matrix`)	design matrix Z, covariance C, eigenvectors	contiguous storage for the Jacobi loops
Ordered map (`map`)	frequency tables and annual climate series	sorted output without extra code
Vector of vectors	bootstrap replicates, per-cluster observation groups in the likelihood	variable-length groups
Queue (`deque`)	the audit log shown in the menu	bounded FIFO

The central relationships are shown below.

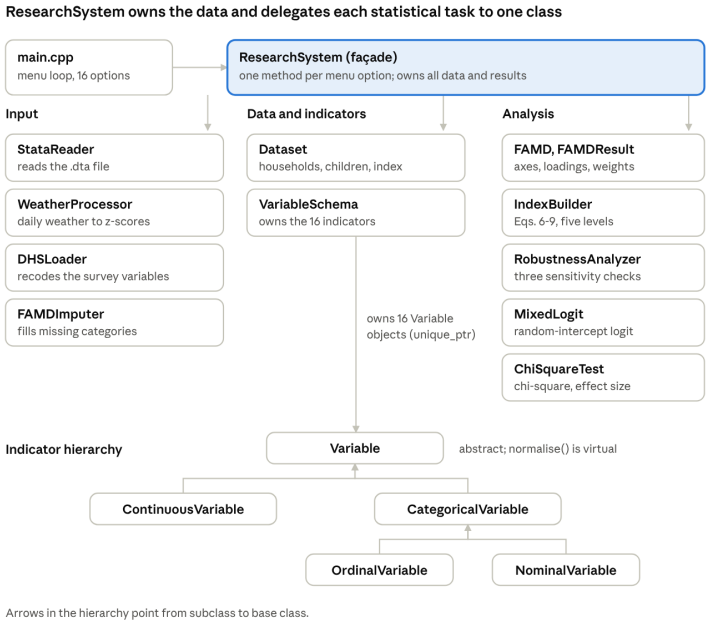


Figure A. Class relationships; the variable hierarchy at the bottom is the polymorphic core.

4 Programmer documentation

This section explains how the program runs and how another programmer can extend it.

4.1 How the program runs

`main.cpp` shows a menu of 16 options (data, the three objectives, tools) and calls one method of `ResearchSystem` per choice. Results are computed lazily and cached: option 6 triggers `ensureData()`, `ensureFAMD()` and `ensureIndex()` in turn if they have not run. Every method writes to a `std::ostream`, so the same code prints to the screen and to the exported report (option 15).

Command-line switches `-auto` and `-selftest` run the whole pipeline or the tests without the menu.

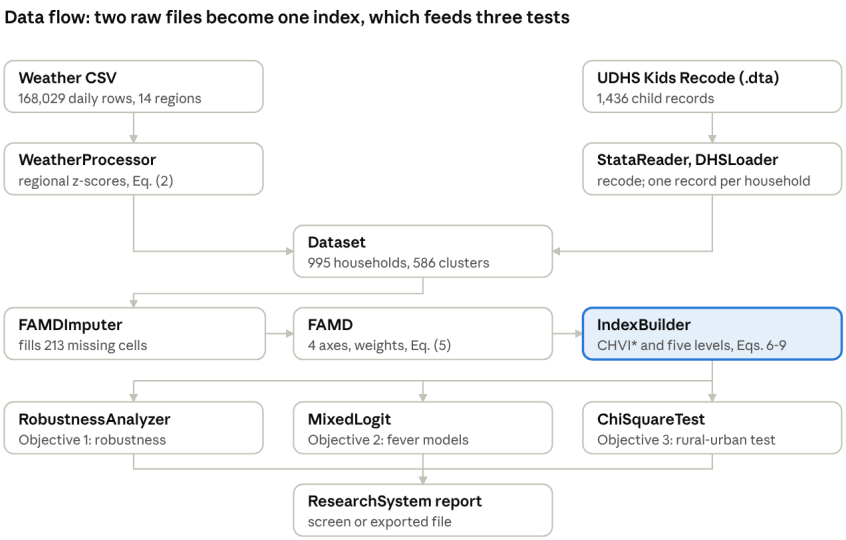


Figure B. Data flow of one run of the complete pipeline (menu option 12).

4.2 Key algorithms

Algorithm 1: climate anomalies (`WeatherProcessor::process`)

```
for each row of the weather file:
    skip if the whole line was seen before            (hash set)
    add to accumulator[region, day]                   (hash map: mean T, H, W; sum of precipitation)
for each accumulator: add to annual[region, year]     (ordered map)
for each region: baseline = values of 1991..2020; recent = mean of 2021, 2022
    z = (recent - mean(baseline)) / sd(baseline)
```

Algorithm 2: FAMD, weights and index (`FAMD::fit`, `FAMDResult::weights`, `IndexBuilder::build`)

```
build Z (n x J): standardise continuous; variable columns (1[x=c] - p_c) / sqrt(p_c)
C = Z'Z / n ; (lambda, V) = JacobiEigen(C), sorted descending
F = Z V[:, 1..K]
for each variable i, axis k: L[i,k] = r^2 (continuous) or eta^2 (categorical)
W[i] = sum_k L[i,k] * lambda[k] / sum_k lambda[k]
Y = score matrix ; E, S, A = weighted means per domain (Eq. 6)
CHVI = (E + S - A) / 3 ; CHVI* = z-score ; level = upper_bound(cuts, CHVI*)
```

Algorithm 3: imputation (`FAMDImputer::impute`)

```
fill every missing category with the mode
repeat up to 100 times:
    fit FAMD ; reconstruct Z-hat from 2 axes, shrinking each by (lambda - s2) / lambda
    for each missing cell: category = argmax over c of  Z-hat[i,c] * sqrt(p_c) + p_c
    stop when no cell changed
```

Algorithm 4: cluster bootstrap (`RobustnessAnalyzer::clusterBootstrap`)

```
for b = 1..B:
    draw n_clusters cluster ids with replacement ; rows = their households   (hash-map lookup)
    fit FAMD on rows ; weights_b ; score ALL households with weights_b
    rho_b = Spearman(CHVI_primary, CHVI_b)
report median, minimum, maximum of rho
```

Algorithm 5: marginal likelihood of one cluster (`MixedLogit::negLogLik`)

```
eta_i = x_i' beta
find mode z* of  h(z) = sum_i [y_i (eta_i + sigma z) - log(1 + exp(eta_i + sigma z))] - z^2 / 2   (Newton)
s = 1 / sqrt(-h''(z*))
logL = log(sqrt(2) s) + logsumexp_q [ log w_q + t_q^2 + h(z* + sqrt(2) s t_q) ] - 0.5 log(2 pi)
```

The optimiser is BFGS with a backtracking line search over (β, log σ); the Hessian at the optimum gives standard errors.

4.3 How to extend the program

1. **New variable.** Add one `add<...>()` line in `VariableSchema::build()` with its labels and scores, add its recode in `DHSLoader::load()` and add its column index constant. FAMD, imputation, weights and the index adapt automatically because the number of design columns is derived from the schema.
2. **New variable type.** Derive from `Variable`, implement `kind()` and `normalise()` and extend the column building in `FAMD::fit()`.
3. **New fever model.** Add a value to the `Cov` enumeration, give it a name and value in `covName()` and `covValue()` and add a `Spec` entry in `ResearchSystem::feverModels()`.
4. **New data source.** `WeatherProcessor` needs only region, date (m/d/yyyy) and four daily columns; the survey side needs the recodes in `DHSLoader`.
5. **Build.** `make` (C++11 or later, no external libraries); `./chvi -selftest` must still report all checks passed.

5 Testing

All 75 built-in checks pass and the real-data results agree with an independent re-implementation. Testing had four layers: unit checks, checks against independent counts, cross-validation and robustness of the program itself.

5.1 Unit checks (menu option 14)

Group	Checks	Example of what is verified
Linear algebra	5	Jacobi eigenvalues of [[2,1],[1,2]] are 3 and 1; VDV' rebuilds a random 7×7 matrix to 5e-15; inverse(A) times A is I; a singular matrix throws
Statistical functions	13	Gauss–Hermite weights sum to sqrt(pi); chi-square tail at 3.841 (1 df) is 0.05; Spearman with ties equals 8/sqrt(95); OLS recovers y = 1 + 2x
Variable classes	9	Eq. (3) gives 0.25 and Eq. (4) gives 0.75 for x = 2.5 on [0, 10]; an unscored category gives NaN; polymorphic call through a base reference

Group	Checks	Example of what is verified
DHS recoding	6	water code 32 is Unimproved; fuel code 6 is Coal/Lignite; birth weight 9998 is unknown
Weather processing	5	a duplicate row is dropped; duplicate-day precipitation is summed; z equals a hand-computed value
Chi-square test	4	the table [[10,20],[30,40]] gives 0.7937 (hand calculation) and V = 0.0891
Multilevel regression	8	simulated data with β = (-0.5, 0.8) and σ = 0.7 give estimates -0.60, 0.79 and 0.73, inside the 95% interval
Data structures	2	the heap keeps the three largest values in order; the log queue drops the oldest entry
Real data	23	see 5.2

5.2 Real-data checks

Counts were first obtained independently with pandas and then compared with the program: 1,436 observations and 6,356 variables in the .dta file, 426 urban records, 275 blank time-to-water values, wealth counts 382/287/268/247/252 and 352 / 936 / 132 / 16 fever answers (yes / no / blank / don't know). The pipeline outputs were then compared with the values reported in Chapter 4 which were produced in R with FactoMineR and lme4.

Quantity	Chapter 4 (R)	This program
Variance explained by 4 axes	30.6%	30.58%
Weights: wealth, fuel, wall, floor	0.517, 0.377, 0.347, 0.329	0.517, 0.377, 0.347, 0.329
Exposure weights	0.125 to 0.199	0.125 to 0.199
Households per level, Very Low to Very High	77, 252, 310, 307, 49	77, 252, 310, 307, 49
CHVI vs S / A / E (r)	+0.867 / -0.904 / +0.557	+0.867 / -0.904 / +0.557
Spearman ρ, 4 vs 12 axes	0.998	0.998
Rural–urban chi-square (4 df)	279.45	279.446
AIC of M0, M1, M2, M3	1488.7, 1474.7, 1496.3, 1483.0	1488.74, 1474.68, 1496.32, 1483.00

5.3 Robustness of the program

- **Memory and undefined behaviour.** The test suite and full pipeline run without errors under AddressSanitizer and UBSan.
- **Compilers.** The code compiles without warnings in strict C++11 and in C++17 and gives identical reports; this matters because Dev-C++ ships an older MinGW compiler.
- **Bad input.** Non-numeric or out-of-range menu entries are re-prompted; a missing data file produces a message that names the file and what to do.
- **A bug found by testing.** The cross-check of missing values showed that the Stata reader did not treat the missing-value code of double fields as missing. It was invisible with the default recoding but would have mishandled blank time-to-water values under the alternative coding. The comparison was corrected to include the boundary value and a regression check (275 blanks in v115, 248 in birth weight) was added.

6 Example output

The program, run on the real data with its default settings, shows that household vulnerability is driven by wealth and housing, that it predicts child fever and that rural households are far more vulnerable than urban ones. The tables below are taken from the full report (`output/full_report.txt`).

6.1 Data and climate exposure

The 1,436 child records come from 995 households in 586 clusters; 32.1% of households are urban. Fever in the last two weeks was reported for 352 children (24.5%), denied for 936 (65.2%) and missing or "don't know" for 148 (10.3%) which leaves 1,288 children for the models. Standardized climate anomalies vary widely by region. Humidity in 2021–2022 is below its 1991–2020 baseline in 13 of 14 regions (z from -3.8 in Kampala to +0.3 in Bukedi), wind is above baseline in 12 of 14 (up to +3.5 in Buganda) and precipitation is lowest in Kampala, Bugisu and Bunyoro (z close to -2.4).

6.2 FAMD and variable weights

The first four axes have eigenvalues 4.971, 2.191, 1.793 and 1.443 and explain 14.6%, 6.4%, 5.3% and 4.2% of the total inertia of 34, a cumulative 30.6%. The Kaiser rule would keep 12 axes. Wealth, cooking fuel and housing materials carry the largest weights while climate anomalies carry small ones (Figure 1).

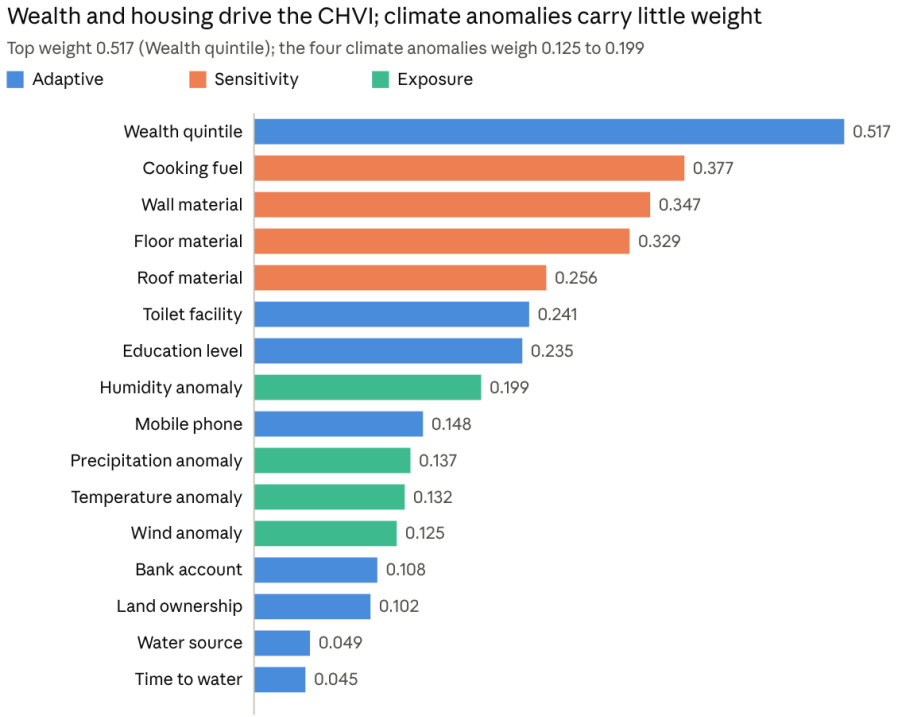


Figure 1. variable weights from Eq. (5) with $K = 4$ axes (UDHS 2022 10% extract with Open-Meteo climate; menu option 5).

6.3 The index and its validity

The statistical properties and construct validity of the index are summarized below.

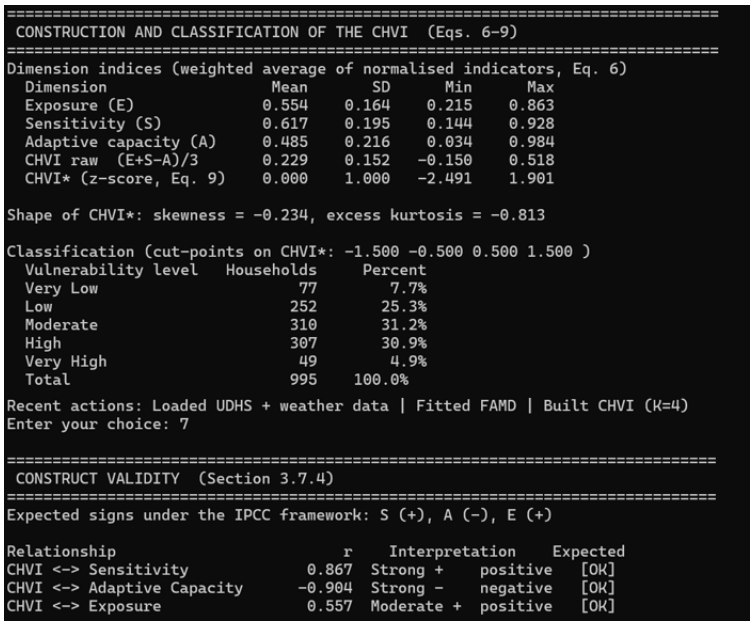


Figure 2. The index and its validity.

CHVI* is slightly flat and left-skewed (skewness -0.23, excess kurtosis -0.81). Its classes are Very Low 77 households (7.7%), Low 252 (25.3%), Moderate 310 (31.2%), High 307 (30.9%) and Very High 49 (4.9%). Construct validity is as the IPCC framework predicts: CHVI* correlates +0.867 with Sensitivity, -0.904 with Adaptive Capacity and +0.557 with Exposure.

6.4 Robustness

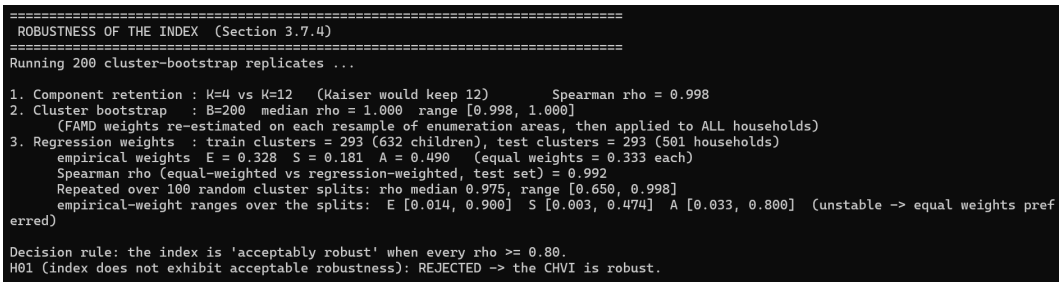


Figure 3. Robustness.

All three median correlations exceed 0.80, so H01 is rejected: the index is acceptably robust.

6.5 CHVI and child fever

```
======================================================================
 MULTILEVEL LOGISTIC MODELS FOR CHILD FEVER  (Eq. 10)
======================================================================
Children with observed fever status (h22 = 1 or 2): 1288  in 586 clusters.  Adaptive Gauss-Hermite quadrature, 1 node (= Laplace appr
oximation, as lme4::glmer default).

Model                                        LogLik       AIC     dAIC    sigma     ICC
M0 Null (random intercept only)             -742.37   1488.74       --     0.94    0.21
M1 Contextual (CHVI + Residence + Radio)    -732.34   1474.68   -14.06     0.88    0.19
M2 Individual (Age+Sex+BirthOrder+BirthWt)  -741.16   1496.32     7.58     0.94    0.21
M3 Full (Contextual + TV + Individual)      -730.50   1483.00    -5.74     0.88    0.19

Preferred specification (lowest AIC): M1 Contextual (CHVI + Residence + Radio)
Likelihood-ratio test M1 vs M0: chi2(3) = 20.06,  p = 1.65e-004

Fixed effects of M1 Contextual (CHVI + Residence + Radio)
  Term                    Beta      SE        z         p       OR       95% CI (OR)
  (Intercept)           -0.960   0.188   -5.094     0.000    0.383      [0.26, 0.55]
  CHVI (z)               0.360   0.094    3.828     0.000    1.434      [1.19, 1.72]
  Urban (vs rural)      -0.039   0.201   -0.194     0.846    0.962      [0.65, 1.43]
  Radio (v158)          -0.112   0.084   -1.329     0.184    0.894      [0.76, 1.05]

CHVI effect in M1: OR per 1-SD increase = 1.434 (p = 1.29e-004)
H02 (CHVI has no association with fever): REJECTED -> CHVI is significantly associated with fever.
```

Figure 4. CHVI and child fever.

M1 has the lowest AIC and improves on the null model (likelihood ratio chi-square 20.06 on 3 df, p = 0.0002). In M1 the odds of fever rise by a factor of 1.43 for each standard-deviation increase in CHVI* (95% CI 1.19 to 1.72, p = 0.0001); residence (OR 0.96, p = 0.85) and radio (OR 0.89, p = 0.18) are not significant. The individual model fits worse than the null and adding television and the individual traits to M1 raises the AIC by 8.3. H02 is rejected.

6.6 Rural–urban disparity and hot spots

The share of households in the High and Very High classes is 46.3% in rural areas and 13.5% in urban areas while 65.5% of urban but only 17.8% of rural households are Very Low or Low (Figure 2). The chi-square statistic is 279.4 on 4 df ($p < 0.0001$) and Cramér's V is 0.53, a large effect, so H03 is rejected. The heap-based listing of the ten most vulnerable households finds all of them in rural Bukedi or Karamoja with CHVI* between 1.72 and 1.90. Ranking regions by mean CHVI* puts Karamoja first and Kampala last (Figure 7).

```
============================================================================
 RURAL-URBAN DISPARITIES IN THE CHVI  (Section 3.7.6)
============================================================================
Distribution of vulnerability level by residence (column %)
  Level                 Urban (%)   Rural (%)   n urban   n rural
  Very Low                   23.2         0.4        74         3
  Low                        42.3        17.3       135       117
  Moderate                   21.0        35.9        67       243
  High                       11.3        40.1        36       271
  Very High                   2.2         6.2         7        42

High + Very High : urban 13.5%  vs rural 46.3%
Very Low + Low   : urban 65.5%  vs rural 17.8%

Pearson chi-square = 279.446,  df = 4,  p = 2.93e-059,  Cramer's V = 0.530
H03 (no rural-urban disparity): REJECTED -> significant rural-urban divide.
```

Figure 5. Rural–urban disparity Chi-square output.

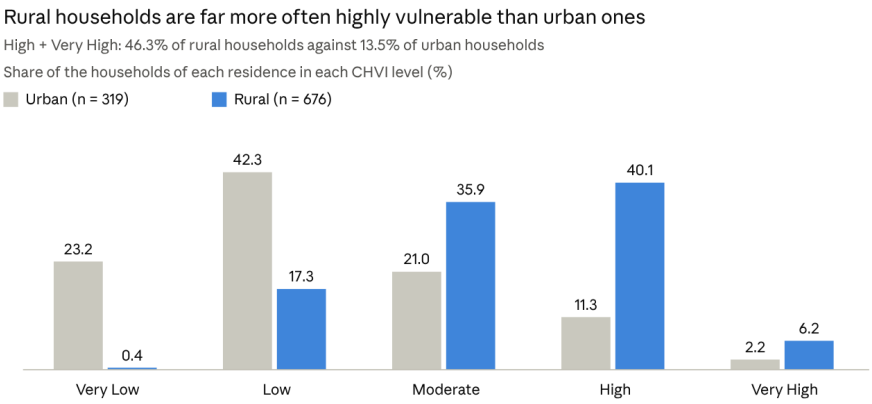


Figure 6. Share of households in each CHVI level by residence: 995 households (319 urban, 676 rural); $\chi^2 = 279.4$ on 4 df, Cramér's V $= 0.53$ (menu option 10).

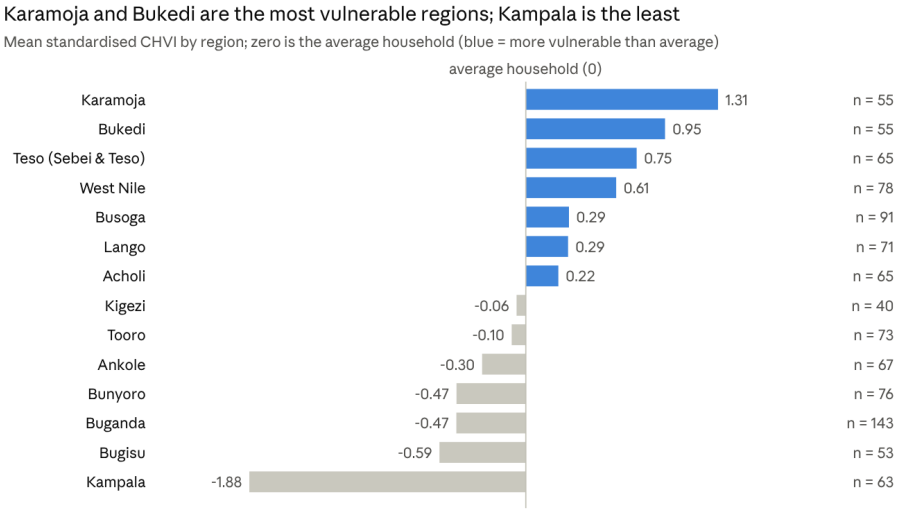


Figure 7. Mean standardized CHVI of the 995 households by region (menu option 11).

7 Critical discussion of the output

The program worked correctly and gives the right answers where a reference exists. Every number in the table of Section 5.2 matches the R results of Chapter 4.

7.1 Where the output differs from Chapter 4 and why

- **Cluster bootstrap.** Chapter 4 reports a median ρ of 0.885. The R code correlates each household's CHVI* with that of the first household of its cluster, so that figure measures similarity within clusters, not stability of the index. The program refits FAMD on every resample and scores all households and obtains ρ of 0.998 to 1.000. This is the stricter and more meaningful test, but it is lenient in one respect, only the weights change while the scoring tables and cut-points are fixed.
- **Regression-weighted CHVI.** Chapter 4 reports weights of 0.019, 0.013 and 0.968 from one split. The program finds that such weights depend on the split: over 100 random splits the Exposure weight ranges from 0.01 to 0.90 and ρ from 0.65 to 0.998. The equal-weighted index is therefore justified. Median ρ (0.975) still exceeds 0.80.

- **Estimation method.** With one quadrature node (the Laplace approximation, as in R) the AICs equal Chapter 4. With 15 nodes the null model has $\sigma = 1.09$ instead of 0.94 and the AIC changes become -12.8 (M1), +7.5 (M2) and -4.5 (M3); the CHVI odds ratio is 1.44 in both. The conclusions do not depend on the choice.

7.2 Accuracy and limits of the inference

Numerical accuracy is high, the Jacobi eigen-solver reproduces matrices to 5×10^{-15} and the quadrature integrates exactly the polynomials tested. The survey is cross-sectional, so the association of CHVI with fever is not proof of cause. Fever is reported by mothers and stands in for malaria which it also overstates. The 10% extract has 995 households and about 2.4 children per cluster which makes the random-effect variance imprecise and the Laplace approximation less exact.

The effect is also modest one standard deviation of CHVI* raises the odds of fever by about 43% and the AIC gain over the null model is 14. Because Sensitivity and Adaptive capacity dominate, the index mostly measures poverty and housing quality.

8 Conclusions and recommendations

The C++ system builds a valid, robust household CHVI from real survey and climate data. Its main findings are:

- **Objective 1.** Wealth, cooking fuel and housing materials drive the index; climate anomalies carry small weights. The index has the expected correlations (+0.867 with Sensitivity, -0.904 with Adaptive capacity, +0.557 with Exposure) and is stable across the number of axes, bootstrap resamples and weighting schemes. H01 is rejected.
- **Objective 2.** One standard deviation of CHVI* raises the odds of child fever by 43% (95% CI 1.19 to 1.72). The contextual model fits best. H02 is rejected.
- **Objective 3.** High and Very High vulnerability affects 46.3% of rural and 13.5% of urban households (chi-square 279.4, 4 df, Cramér's V = 0.53). H03 is rejected.
- **Programming.** The OOP design (polymorphic indicators, a façade class, templates and STL structures) keeps each statistical task in one class and 75 self-tests plus an independent cross-check protect against regressions.

Recommendations.

1. Use of more than 50% of UDHS dataset to capture most of the population-level variation or regional disparities.
2. Link exposure to household GPS coordinates which would give the Exposure dimension real within-region variation.

References

Abdrabo, K. I., Hassaan, M. A., El-Barmelgy, M. M. and Zahran, A. (2023). An integrated indicator-based approach for assessing urban climate vulnerability. *Urban Climate*, 48:101426.

Conlon, K. C., Kintziger, K. W., Jagai, J. S., Stone, M. V., Hess, J. J. and Uejio, C. K. (2020). Mapping human vulnerability to extreme heat: a critical assessment of heat vulnerability indices created using principal components analysis. *Environmental Health Perspectives*, 128(9):097001.

Cutter, S. L., Boruff, B. J. and Shirley, W. L. (2003). Social vulnerability to environmental hazards. *Social Science Quarterly*, 84(2):242–261.

Erima, G., Bamutaze, Y., Gidudu, A., Egeru, A. and Kabenge, I. (2025). Dimensions and drivers of social vulnerability to flood risk in Manafwa catchment, eastern Uganda. *Cogent Social Sciences*, 11(1):2493353.

Gbaguidi, G. J., Topanou, N., Walter, L. F. and Ketoh, G. K. (2025). Assessing vulnerability to malaria in the context of climate change: a comprehensive study in northern Benin, West Africa. *Discover Social Science and Health*, 5(1):30.

Husson, F., Josse, J. and Pagès, J. (2010). Principal component methods, hierarchical clustering, partitional clustering: why would we need to choose for visualizing data? Technical report, Agrocampus Rennes.

Isunju, J. B., Bukenya, J. N., Masembe, V., Ssekamatte, T. and Namanya, D. B. (2024). Climate change health vulnerability and adaptation assessment for sound management of climate change-related health risks in Uganda. Ministry of Health Uganda and Makerere University School of Public Health, Kampala.

Isunju, J. B., Magala, J., Wafula, S. T., et al. (2026). Household vulnerability to climate impacts on WASH and risk mitigation in informal settlements of eastern Uganda. *Scientific Reports*, 16(1):21154.

Lê, S., Josse, J. and Husson, F. (2008). FactoMineR: an R package for multivariate analysis. *Journal of Statistical Software*, 25(1):1–18.

López-Carr, D., Mwenda, K. M., Pricope, N. G., et al. (2016). Climate-related child undernutrition in the Lake Victoria Basin: an integrated spatial analysis of health surveys, NDVI and precipitation data. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 9(6):2830–2835.

Obasohan, P. E., Walters, S. J., Jacques, R. and Khatab, K. (2023). Risk factors associated with multimorbidity among children aged under-five years in sub-Saharan African countries: a scoping review. *International Journal of Environmental Research and Public Health*, 20(2):1377.

Pagès, J. (2004). Multiple factor analysis: main features and application to sensory data. *REVSTAT–Statistical Journal*, 2(1):43–64.

Uganda Bureau of Statistics and ICF (2023). *Uganda Demographic and Health Survey 2022: Key Indicators Report*. UBOS, Kampala and ICF, Rockville.

van de Ven, G. W., de Wit, A. J. and van Wijk, M. T. (2019). Vulnerability and adaptation options to climate change for rural livelihoods in Uganda. *Agricultural Systems*, 176:102661.

Vincent, K. (2004). Creating an index of social vulnerability to climate change for Africa. Working Paper 56, Tyndall Centre for Climate Change Research.

Wang, P., O'Brien, F., Son, J. Y., et al. (2026). An updated modeling framework and sensitivity analysis of methodology for the climate health vulnerability index. *Nature Communications*, 17(1):68162.

Zhang, R., Liu, Y. and Wang, X. (2025). Exposure-sensitivity-adaptability (ESA) framework for disaster vulnerability modeling. *Ecological Indicators*, 176:113662.

Zippenfenig, P. (2024). Open-Meteo climate API. Zenodo.

Appendix

A. Source codes submitted with the report

File	Lines	Contents
`main.cpp`	189	menu, settings, command-line options
`ResearchSystem.hpp`	585	façade class, all analysis stages and printing
`SelfTest.hpp`	389	75 verification checks
`Variable.hpp`	261	indicator class hierarchy and schema
`MixedLogit.hpp`	261	random-intercept logistic regression
`FAMD.hpp`	226	FAMD, loadings, weights
`Stats.hpp`	217	statistical functions, quadrature, RNG, OLS
`DHSLoader.hpp`	196	survey recoding
`Robustness.hpp`	164	three robustness analyses
`Matrix.hpp`	161	matrix, inverse, Jacobi eigen-solver
`StataReader.hpp`	145	.dta reader
`Climate.hpp`	144	weather processing, Eq. (2)
`Index.hpp`	142	index construction, classifier
`Data.hpp`	116	records and dataset
`Imputer.hpp`	96	FAMD imputation
`DataStructures.hpp`	67	min-heap, log queue
`Disparity.hpp`	61	chi-square test
`Portable.hpp`	54	portability helpers

B. Test data and output files

`output/full_report.txt` (complete run), `output/selftest_output.txt` (75 checks), `output/oracle_check_output.txt` (independent cross-check), `output/sensitivity_blank_water_on_premises.txt` (alternative coding), `output/households_chvi.csv` (household results).

C. Program menu

```
Climate-Health Vulnerability Index for Uganda
Author : Kuli Faith Rombe (2025/HD06/22009U)
OOP Programming and Data Structures Implementation

+-----------------------------------------------------------------+
|  CLIMATE-HEALTH VULNERABILITY INDEX (CHVI) FOR UGANDA           |
|  UDHS 2022 + Open-Meteo climate | FAMD | multilevel logit | chi2 |
+-----------------------------------------------------------------+
|  DATA                                                           |
|   1. Load UDHS (.dta) and weather (.csv) data                   |
|   2. Dataset summary (fever prevalence, residence)              |
|   3. Climate anomalies by region (Eq. 2)                        |
|   4. Recoding, missing data and imputation                      |
|  OBJECTIVE 1 : CONSTRUCT THE CHVI                               |
|   5. Factor Analysis of Mixed Data and indicator weights        |
|   6. Construct and classify the CHVI                           |
|   7. Construct validity                                         |
|   8. Robustness checks (components, bootstrap, regression)     |
|  OBJECTIVE 2 : ASSOCIATION WITH CHILD FEVER                     |
|   9. Multilevel logistic regression models (M0-M3)              |
|  OBJECTIVE 3 : RESIDENCE DISPARITIES                            |
|  10. Rural-urban chi-square test                                |
|  TOOLS                                                          |
|  11. Most vulnerable households and regional ranking            |
|  12. Run the complete pipeline                                  |
|  13. Settings (files, components, bootstrap size, top-k)        |
|  14. Run self-tests                                             |
|  15. Export full report to file                                 |
|  16. Export household-level results (CSV)                       |
|   0. Exit                                                       |
+-----------------------------------------------------------------+
```

Figure 4. Program menu.